Lecture 29: Introduction to Digital Logic Families; classification of Digital Logic Families; Characteristics of Digital ICs.

Miniature, low-cost electronics circuits whose components are fabricated on a single, continuous piece of semiconductor material to perform a high-level function. This IC is usually referred to as a monolithic IC first introduced in 1958. The digital ICs are categorized as,

1. **Small scale integration or SSI (<12 no of gates)**

2. **Medium scale integration or MSI (12 to 99 no of gates)**

3. **Large scale integration or LSI (100 to 9999 no of gates)**

4. **Very large scale integration or VLSI (10,000 or more)**

Characteristics of Digital ICs

@Input /Output voltage level:

V_{OHmin} : The minimum output voltage in HIGH state (logic '1'). V_{OHmin} is 2.4 V for TTL and 4.9 V for CMOS.

V_{OLmax} : The maximum output voltage in LOW state (logic '0'). V_{OLmax} is 0.4 V for TTL and 0.1 V for CMOS.

V_{IHmin} : The minimum input voltage guaranteed to be recognised as logic 1. V_{IHmin} is 2 V for TTL and 3.5 V for CMOS.

V_{ILmax} : The maximum input voltage guaranteed to be recognised as logic 0. V_{ILmax} is 0.8 V for TTL and 1.5 V for CMOS.

@Fan-in

Fan-in (input load factor) is the number of input signals that can be connected to a gate without causing it to operate outside its intended operating range. expressed in terms of standard inputs or units loads (ULs).

@Fan-out

Fan-out (output load factor) is the maximum number of inputs that can be driven by a logic gate. A fan-out of 10 means that 10 unit loads can be driven by the gate while still maintaining the output voltage within specifications for logic levels 0 and 1.

@Propagation Delay:

The time required for the output of a digital circuit to change states after a change at one or more of its inputs. The speed of a digital circuit is specified in terms of the propagation delay

time. The delay times are measured between the 50 percent voltage levels of input and output waveforms. There are two delay times, t_{pHL}: when the output goes from the HIGH state to the LOW state and t_{pLH}, corresponding to the output making a transition from the LOW state to the HIGH state. The propagation delay time of the logic gate is taken as the average of these two delay times.

@Noise Margin

LNM (Low noise margin): The largest noise amplitude that is guaranteed not to change the output voltage level when superimposed on the input voltage of the logic gate (when this voltage is in the LOW interval). $LNM=V_{ILmax}-V_{OLmax.}$

HNM (High noise margin): The largest noise amplitude that is guaranteed not to change the output voltage level if superimposed on the input voltage of the logic gate (when this voltage is in the HIGH interval). $HNM=V_{OHmin}-V_{IHmin}$

@Threshold Voltage:

The threshold voltage is defined as that voltage at the input of a gate which causes a change in the state of the output from one logic level to another.

@Power Dissipation:

The Power Dissipation of a logic gate is the supply power required by the logic gate to operate with 50 percent duty cycle at a specified operating frequency.

@Figure of Merit:

The speed power product is called figure-of-merit of the IC family. It is the product of the propagation delay and power dissipation of the gate.

The basic Classification of the Logic Families are as follows:

A) Bipolar Families
B) MOS Families
C) Hybrid Devices

A) Bipolar Families:
1. Diode Logic (DL)
2. Resistor Transistor Logic (RTL)
3. Diode Transistor Logic (DTL)
4. Transistor- Transistor Logic (TTL)
5. Emitter Coupled Logic (ECL) or Current Mode Logic (CML)
6. Integrated Injection Logic (IIL).

B) MOS Families:
1. P-MOS Family
2. N-MOS Family

3. Complementary-MOS Family

Among complementary-MOS family

i)Standard C-MOS
ii)Clocked C-MOS
iii)Bi-CMOS
iv)Pseudo N-MOS
v)C-MOS Domino Logic
vi)Pass Transistor Logic

Hybrid Family:
Bi-CMOS Family

Families of Integrated Circuits(Brief):

A logic family is a group of electronic logic gates. A family has its own discrete logic level, power supply, individual components, characteristic, advantages and disadvantages. Within each family, there is a range of voltage which may be low level or high level.

These families are listed as following below:

Diode Logic (DL)

Registers and diodes are used to implement logic. The purpose of diode is to perform OR and AND operation, and logic switch. It is essential that diode is forward biased so that it can conduct. Disadvantages: diodes can not perform NOT operation, diode cannot work for multiple states, only one stage at a time, tend to degrade signals quickly.

Resistor-Transistor Logic (RTL)

Registers and transistors are used to implement logic. Transistors are used to combine signals which are inverted and amplified inputs. These are economical and easy to design but slower in speed. These require great amount of current from supply. RTL gates can be used as interface between linear circuits and digital.

Diode Transistor Logic (DTL)

Registers and diodes are used to implement logic. It has advantages over DL and RTL. Diode can perform AND and OR operation along with a transistor and output signal can be amplified. It has logic inversion which is signal can be restored to full logic level, if we add a transistor at output of logic gates. OR operation can be performed using diodes instead of resistors which removes interaction between input signals. Since input resistor to transistor, switching speed of transistor is limited. DTL was used in tube computers.

Transistor - Transistor Logic (TTL)

Transistors are used to implement logic and Bipolar transistors to contract integrated circuits. There are standard TTL, high speed TTL, low power TTL, and schottky TTL. It is popular logic family.

Emitter Coupled Logic (ECL)

ECL is a non-structured logic. Advantages of this logic are speed, and very low propagation delay.

Complementary Metal Oxide Semiconductor Logic (CMOS)

It is known for high fan-out, low power consumption and most reliable logic family.

----------END OF LECTURE----------

REFERENCES

1. **Text book**: **T1.** Modern Digital Electronics by R. P. Jain (Tata McGraw Hill)
2**. Reference book**: **R1.** Fundamental of digital circuits by A. Anand Kumar (PHI)

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